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Introduction

The purpose of this report is to present a conceptual guideline for my future studying website. This provides the outlines of the initial objective of designing a few study techniques. By articulating the structural features, this will provide the readers with my ideologies and methods in order to enhance my own education. This initial iteration in which research-based study techniques are discussed and then implemented into my own education in order to understand the detail needed for the website.

What are Study Techniques

Study techniques are structured studying approaches that are designed to improve retention in a certain topic. Some of the most used and effective (when done right) are Spaced Repetition, Active recall, etc. For this intended project, my goal is to find the best study techniques and converge into a mix of techniques that will work hand in hand with one another.

Why are Study Techniques Important?

For students, 9th graders specifically, this transition into high school is a critical moment. Academic expectations increase significantly, and students who develop strong study habits early gain a lasting advantage.

By watching videos by Andrew Huberman and Justin Sung, I have noticed a few major things when they discuss study habits/ techniques. Andrew usually connotes that proper studying correlates to the brain’s neuroplasticity. While Justin Sung views studying as **note memorization or passive reading, but as actively *processing* information.** This contrast shows how important it is to understand how to study correctly rather than simply spending more time doing it. It also helped me realize that reading isn’t just getting through text, but it is to engage with it and strengthen your understanding. This is why we must look at how to study correctly.

According to a report done by Stanford University, many students fall short academically because they did not employ a strategic approach for their learning. Even some of the professors, most bright students, fell short due to the simple reason that they didn’t know how to read purposefully. The article’s author, Patrica Chen, said: “Strategize how you want to effectively direct your efforts before you pour your energy into it. ”

Some Techniques and their differences

For this section, I have found some of the most useful and effective study techniques and placed them on this table. There are 4 sections that will decide their importance for 9th graders. This table will show the good and the badness of each technique and see if they will provide anything meaningful. This is a list of the types that are most effective:

Study Techniques	Curricular Transferability	Shows What Don't You Know?	Stops Brain Freeze?	Easy to Alone?	Is It Good for 9th Graders?
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Active Recall	Yes. Great for all subjects, especially science and math.	Shows how much the reader processes information.	Yes. It helps students to focus on one small topic at a time and highly prevents panic	It can be done alone but it is better to have notes or a book with you	PubMed and science directly say, it is beneficial for students by providing confidence in the topic.
Spaced Repetition	Yes. Keeps math and science-based topics fresh even in prolonged periods	Not Really, focuses on stopping forgetting rather than finding mistakes.	Perfect for this, studying over days eliminates test cramming.	Hard. Requires strict tracking and needs a lot of information to look back to	Hard to build as a 4-15-year-old. It involves planning weeks in advance.
Fenman Technique	Yes. Especially good for harder topics in science.	High. Trying to explain it simply catches hidden weak spots instantly.	Hard but doable, requires a lot of upfront brainpowers to simplify terms.	You just need to speak aloud or write a summary.	Highly effective for 9th graders, helps them move past form memorization. However, it takes a bit of time.
SQ3R	Yes.	Stops you from passively skimming pages without understanding them.	Yes, it needs to follow a strict process where each step requires	Yes, it is designed as a self-directed tool without teachers or others.	It is highly effective for 9th graders due to their shift into academic journeys.

Issues of each Technique

To continue forward, we must understand the flaws in each of these four major techniques. This will paint a holistic picture of their flaws and their positive attributes in order to help the reader and myself to understand which ones are going to be the best for me.

- Active Recall:

Active recall requires intense cognitive effort, making it highly exhausting for students. Creating high quality self testing questions also takes a substantial amount of time. Additionally, a medical review by Osmosis Learning notes that memorizing details can mask conceptual gaps.

(<https://www.osmosis.org/blog/active-recall-the-most-effective-high-yield-learning-technique>)

- Spaced Repetition:

This demands strict, unyielding daily consistency from learners. Missing scheduled review sessions quickly creates overwhelming backlogs of unread material. Neuroscience research from the eLearning Industry proves that this method is ineffective for periods where learners have to understand a topic in a short period of time.

(<https://elearningindustry.com/spaced-learning-neuroscience-based-approach-to-foster-knowledge-retention>)

- Feynman Technique:

While explaining concepts in simple terms requires a high upfront time commitment, this minor drawback is far from a dealbreaker because it forces an unparalleled level of long-term comprehension. An academic study guide from the University of Colorado Boulder demonstrates that teaching concepts out loud is the single best way to expose hidden gaps in your knowledge. This active translation process quickly transforms any initial weak spots into areas of absolute academic strength.

(<https://www.colorado.edu/artssciences-advising/resource-library/life-skills/the-feynman-technique-in-academic-coaching>)

- SQ3R:

Although the comprehensive 5 step system slows down initial reading speeds, this small hurdle is easily managed since it completely eliminates the need for future rereading. According to the efficient reading resources published by Harvard University, this structured method serves as the ideal framework for navigating complex textbooks and dense scientific articles. By prioritizing comprehension over speed, it acts as an incredibly powerful tool for turning passive reading habits into massive grade improvements. (<https://academicresourcecenter.harvard.edu/2024/03/27/sq3r/>)

Books- Why are they important

Books remain one of the most effective methods of information for deep learning. Unlike videos, flashcards, or other sources, books demand extreme dedication and focus to fully process the given information. For 9th graders, developing a reading habit can significantly expand vocabulary, enhance comprehension, and strengthen analytical skills.

- Atomic Habits (By James Clear)

Focus of the book: The main point of *Atomic Habits* by James Clear is that remarkable results come from the compound effect of small, consistent daily improvements

Comments: Atomic is not a book purely for academic study but teaches people (especially teens) how to execute those techniques consistently and shows students how to change their environment that makes studying smooth by using Habit Stacking.

- Make It Stick

Author: Peter C. Brown, Henry L. Roediger III, & Mark A. McDaniel

Focus of the book: It dismantles ineffective habits like re-reading notes or highlighting text. Instead, it teaches students how to use retrieval practice (testing yourself) and spaced repetition (spacing out study sessions over time) to lock information into long-term memory.

Comments: While Atomic habits teach students how to build a constant routine, Make It Stick teaches them what to do during those study sessions to maximize brain usage. It introduces the concept of "desirable difficulties," showing 9th graders that studying should feel effortful.

- Pragmatic Thinking and Learning: Refactor Your Wetware

Author: Andy Hunt

Focus of the book: It provides a systematic methodology known as the "Dreyfus Model" to completely identify one's exact skill level, (from Beginner to Expert), so you stop using wrong study strategies. It also gives concrete tools to intentionally balance your logical left brain with your creative right brain to master complex technical concepts faster and solve difficult problems.

Videos and Youtubers: Their credentials, teachings, etc.

Video content has become one of the most accessible ways for students to learn about studying. I have provided a list of my favorite educators who have built strong reputations for evidence-based advice on studying, productivity, and correlations of neuroscience/neuroplasticity.

1. Justin Sung: Justin Sung is an educational youtuber focused specifically on efficient studying. His core message that heavily challenges studying harder or longer is the solution for studying. Instead, he advocates active processing techniques like the Feynman Method and retrieval practice. His videos are practical, concise, and directly relevant to the 9th grade experience.
(<https://www.youtube.com/@JustinSung>)
 2. Andrew Huberman: Dr. Andrew Huberman is a professor of neurobiology at Stanford School of Medicine. His podcast/ YouTube channel, Huberman Lab, covers the function, focus, and performance of the brain while studying. He is also the “ambassador” for neuroplasticity, where he frequently talks about how to enhance the brain’s lifelong ability to reorganize, adapt, and create new neural connections. This is the main basis of his ideologies when it comes to studying where he believes neuroplasticity is the deciding factor of a student who performs better than others with his recommendations like active recollection and interleaving.
(<https://www.youtube.com/@hubermanlab>)
 3. Ali Adbaal: doctor-turned YouTube educator who built his following by sharing evidence-based study strategies during his time at Cambridge. His videos cover active recall, spaced repetition, and time management in a straightforward and relatable way. His content is especially useful for students building a study system for the first time. (<https://www.youtube.com/@aliabdaal>)
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The Best Technique For Me

After thoroughly analyzing the four study techniques presented in this specific report, I have concluded that a combination of SQ3R and the Feynman Technique will provide me with the most important and effective method of studying. Of course, Active recall and Spaced repetition is key to academic success so I will incorporate hints of them in order to maintain full understanding.

Why this combination?

These two techniques complement each other in a way that covers the entire learning method. SQ3R is a structured reading method that ensures I never passively skim through material, it forces me to survey, question, read, review, and recite before moving on. As well as it forces me to follow a proper procedure for my learning where it cancels out active recollection and spaced repetition with the addition of the Feynman Technique. By forcing myself to explain concepts in simple terms it will allow me to actively recollect information while also helping myself understand the nuances of the topic. Since this is seeping into grade 9, this will provide the optimal techniques necessary

What do these techniques cover

SQ3R ensures I never passively absorb information, while the Feynman Technique confirms I actually understand it as if I were explaining the concept to a 5 year old. Together, they address two of the biggest problems students face: skimming without comprehension, and memorizing without truly understanding.

Implementation

In order to implement this style of studying, it is best to modify the SQ3R methodology in order to fit the benefits of the Feynman technique.

Here is an example that fits both Math and Science:

Each study session is going to be composed of a 5 minute skim/ survey of the material and thinking about questions that will guide the topic and create a purpose for the session. For example, If I am about to start a chapter in math, I would:

- ☐ Check the topic, definition, guiding factors (like themes, questions,etc.)
- ☐ Then I would think about the main points of the chapter
- ☐ Create at least 2-5 questions to research later

Afterwards I would begin reading the chapter one section at a time instead of trying to finish the entire lesson in one sitting (whatever was planned). As I read and do the given problems, I will find attempt to answer my questions from the survey phase and take notes briefly such as:

- Definitions
- Key problems { any hard or interesting questions (to review back)}
- Formulas

I will also come back to any problem sets or topics that I feel wonky with. Then I will review the Feynman technique by explaining to an imaginary audience, where I will try to explain the topic to various people (five year olds, olds people, etc.). I would aim towards explaining the topic through the lens of the audience (imaginary) and attempt to teach them the key concepts, examples, etc. During this, I will understand the gaps in my reading. And through a recurring process of teaching and active recollection, I should be able to review it properly.

Finally, in order to implement the Recite phase of the SQ3R method, I will set up a buffer window of 1-2 weeks. This will occur after “completing” the topic, when I come back and will explain the topic as if I didn’t know any previous knowledge. The reason I set up a 1-2 weeks buffer zone is to ensure that I would not remain on that topic. It would be more fresh in my memory if I recently did it (say 2 days before reciting) but a gap of that length will help me keep my true understanding. During that period of time it is best to adopt interleaving where I will move on to the next math topic and review back to the previous topic afterwards.

Once doing so, any ideas that I cannot explain clearly will be identified as knowledge gaps and reviewed before continuing onto that “current”/ the topic after reviewing the topic.

Extra: During the review phase or recite, it is also beneficial to review through testing where one creates a test for a certain topic. This will show the flaws in one’s application of the topic rather than Feynman’s Technique where it only shows the conceptual flaws. It may seem old

school, however, testing builds students rigor and also shows the real application of the topic learnt and your ability in it.

This process will be made for 2-3 chapters for my own experimentation while this occurs and slowly my evidence of it working or not will be explained further.

Which methods are best for Notes:

Notes Method	Best For	Fits Techniques	Rating
Cornell Notes ¹	Science, Math, History, Etc.	Yes	Best (Top Choice)
Mind mapping ²	Connecting big ideas before reading	Fits SQ3R Well but partially effective for Feynman Technique	Great (when done correctly)
Outline Method ³	Math, structured chapters	Fits SQ3R Well but partially effective for Feynman Technique	Good
Sentence Method ⁴	Fast Lecture, quick chapters	Not great from each	Good

Sources:

¹ <https://lsc.cornell.edu/how-to-study/taking-notes/cornell-note-taking-system/>

² <https://adelaide.edu.au/students/student-services/academic-support/>

³ https://www.mcla.edu/_pdfs/student-services/academic-support/outliningmethodfornotetaking.pdf

⁴ <https://academicresourcecenter.harvard.edu/2023/10/02/note-taking/>

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